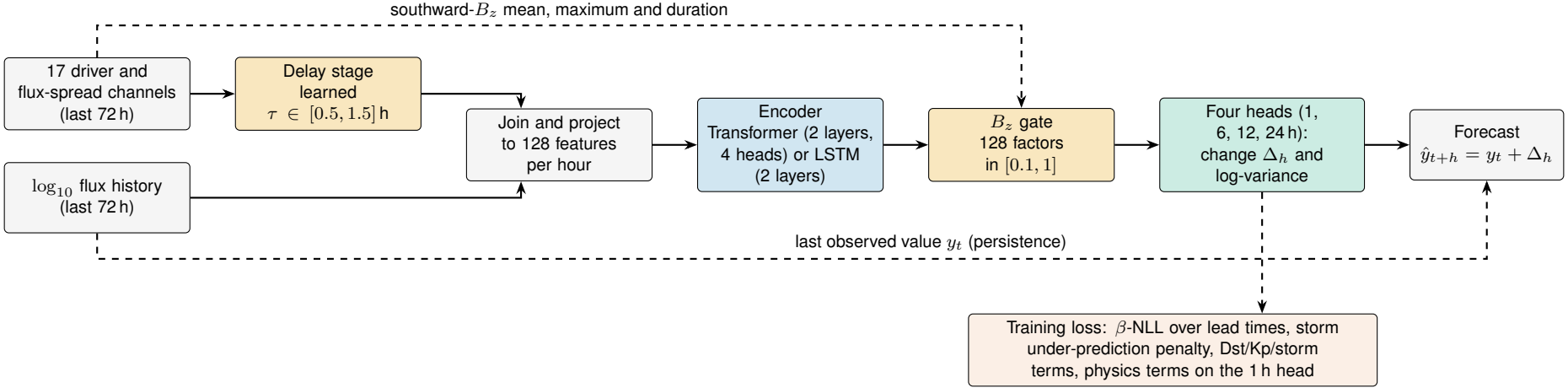
southward-B_z mean, maximum and duration
17 driver and flux-spread channels (last 72 h)
Delay stage learned $\tau \in [0.5, 1.5]$ h
$\log_{10}$ flux history (last 72 h)
Join and project to 128 features per hour
Encoder Transformer (2 layers, 4 heads) or LSTM (2 layers)
B_z gate 128 factors in $[0.1, 1]$
Four heads (1, 6, 12, 24 h): change Δ_h and log-variance
Forecast $\hat{y}_{t+h} = y_t + \Delta_h$
last observed value y_t (persistence)
Training loss: β-NLL over lead times, storm under-prediction penalty, Dst/Kp/storm terms, physics terms on the 1 h head